# Australian Election Trends Analysis (2022–2025)

Shravani Jadhav

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## Overview

This project investigates the shift in Australian federal election voting patterns between 2022 and 2025, following the reported change where minor parties and independents collectively received more first preference votes than the Liberal–National Coalition in the 2025 federal election. The analysis examines whether this national voting shift is reflected across individual electorates and explores the demographic characteristics of electorates experiencing strong changes in voter preferences.

## Fact checker

```
## Load and combine the data

# Load libraries
library(tidyverse)
library(sf)

# Read in 2022 data, skipping the first row
data_2022 <- read_csv("data/HouseDopByDivisionDownload-27966_2022.csv",
                      skip = 1) |>
  mutate(year = 2022)

# Read in 2025 data
data_2025 <- read_csv("data/HouseDopByDivisionDownload-31496_2025.csv",
                      skip = 1) |>
  mutate(year = 2025)

# Combine into one data frame
data_combined <- bind_rows(data_2022, data_2025)

# Check it looks right
glimpse(data_combined)
```

```
Rows: 65,984
Columns: 15
$ StateAb        <chr> "ACT", "ACT", "ACT", "ACT", "ACT", "ACT", "ACT", "ACT~
$ DivisionID     <dbl> 318, 318, 318, 318, 318, 318, 318, 318, 318, 318, 318~
$ DivisionNm     <chr> "Bean", "Bean", "Bean", "Bean", "Bean", "Bean", "Bean~
$ CountNumber    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
```

```
$ BallotPosition    <dbl> 1, 1, 1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 4, 4, 4, 4, 5, 5,~
$ CandidateID       <dbl> 36239, 36239, 36239, 36239, 37455, 37455, 37455, 3745~
$ Surname           <chr> "CONWAY", "CONWAY", "CONWAY", "CONWAY", "AMBARD", "AM~
$ GivenNm           <chr> "Sean", "Sean", "Sean", "Sean", "Benjamin", "Benjamin~
$ PartyAb           <chr> "UAPP", "UAPP", "UAPP", "UAPP", "ON", "ON", "ON", "ON~
$ PartyNm           <chr> "United Australia Party", "United Australia Party", "~
$ Elected           <chr> "N", "N", "N", "N", "N", "N", "N", "N", "Y", "Y", "Y"~
$ HistoricElected   <chr> "N", "N", "N", "N", "N", "N", "N", "N", "Y", "Y", "Y"~
$ CalculationType   <chr> "Preference Count", "Preference Percent", "Transfer C~
$ CalculationValue  <dbl> 2831.00, 2.88, 0.00, 0.00, 2680.00, 2.72, 0.00, 0.00,~
$ year              <dbl> 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022,~
```

```
# See all unique parties
data_combined |>
  distinct(PartyAb, PartyNm) |>
  arrange(PartyAb) |>
  print(n = 100)
```

```
# A tibble: 55 x 2
   PartyAb PartyNm
   <chr>   <chr>
 1 AJP     Animal Justice Party
 2 AJP     AJP
 3 ALP     Australian Labor Party
 4 ALP     Labor
 5 ALP     A.L.P.
 6 ALP     Australian Labor Party (Northern Territory) Branch
 7 ASP     Shooters, Fishers and Farmers Party
 8 AUC     Australian Christians
 9 AUD     Australian Democrats
10 AUP     Australian Progressives
11 AUVA    Australian Values Party
12 CEC     Citizens Party
13 CLP     NT CLP
14 CYA     Australian Federation Party
15 CYA     Trumpet of Patriots
16 DHJP    Derryn Hinch's Justice Party
17 DPDA    Democratic Alliance
18 DPDA    Drew Pavlou Democratic Alliance
19 FFPA    Family First
20 GAP     The Great Australian Party
21 GRN     The Greens
```

```
22 GRN      Queensland Greens
23 GRN      The Greens (WA)
24 GRN      Australian Greens
25 GRN      The Greens (WA) Inc
26 GRPF     Gerard Rennick People First
27 GVIC     The Greens
28 HMP      Legalise Cannabis Australia
29 HMP      Legalise Cannabis Party
30 IAP      Indigenous - Aboriginal Party of Australia
31 IMO      Informed Medical Options Party
32 IMO      HEART Party
33 IND      Independent
34 JLN      Jacqui Lambie Network
35 KAP      Katter's Australian Party (KAP)
36 LDP      Liberal Democrats
37 LNP      Liberal National Party of Queensland
38 LNP      LNP
39 LP       Liberal
40 LTP      Libertarian
41 NP       The Nationals
42 NP       National Party
43 ON       Pauline Hanson's One Nation
44 REAS     Reason Australia
45 SAL      Socialist Alliance
46 SOPA     FUSION: Science, Pirate, Secular, Climate Emergency
47 SOPA     FUSION | Planet Rescue | Whistleblower Protection | Innovation
48 SPP      Sustainable Australia Party - Stop Overdevelopment / Corruption
49 TLOC     The Local Party
50 TNL      TNL
51 UAPP     United Australia Party
52 VNS      Victorian Socialists
53 WAP      WESTERN AUSTRALIA PARTY
54 XEN      Centre Alliance
55 <NA>     <NA>
```

```r
## Classify parties into groups

data_classified <- data_combined |>
  mutate(
    PartyType = case_when(
      PartyAb == "ALP" ~ "major_labour",
      PartyAb %in% c("LP", "NP", "LNP", "CLP") ~ "major_coal",
```

```r
      TRUE ~ "minor_or_indep"
  )
  )

# Check the classification looks right
data_classified |>
  distinct(PartyAb, PartyNm, PartyType) |>
  arrange(PartyType) |>
  print(n = 100)
```

```
# A tibble: 55 x 3
   PartyAb PartyNm                                                      PartyType
   <chr>   <chr>                                                        <chr>
 1 LP      Liberal                                                      major_co~
 2 NP      The Nationals                                                major_co~
 3 CLP     NT CLP                                                       major_co~
 4 LNP     Liberal National Party of Queensland                         major_co~
 5 NP      National Party                                               major_co~
 6 LNP     LNP                                                          major_co~
 7 ALP     Australian Labor Party                                       major_la~
 8 ALP     Labor                                                        major_la~
 9 ALP     A.L.P.                                                       major_la~
10 ALP     Australian Labor Party (Northern Territory) Branch           major_la~
11 UAPP    United Australia Party                                       minor_or~
12 ON      Pauline Hanson's One Nation                                  minor_or~
13 IND     Independent                                                  minor_or~
14 GRN     The Greens                                                   minor_or~
15 LDP     Liberal Democrats                                            minor_or~
16 <NA>    <NA>                                                         minor_or~
17 SOPA    FUSION: Science, Pirate, Secular, Climate Emergency          minor_or~
18 DPDA    Democratic Alliance                                          minor_or~
19 CYA     Australian Federation Party                                  minor_or~
20 CEC     Citizens Party                                               minor_or~
21 SPP     Sustainable Australia Party - Stop Overdevelopment / Corru~  minor_or~
22 AUD     Australian Democrats                                         minor_or~
23 IMO     Informed Medical Options Party                               minor_or~
24 ASP     Shooters, Fishers and Farmers Party                          minor_or~
25 AJP     Animal Justice Party                                         minor_or~
26 TNL     TNL                                                          minor_or~
27 IAP     Indigenous - Aboriginal Party of Australia                   minor_or~
28 SAL     Socialist Alliance                                           minor_or~
29 GRN     Queensland Greens                                            minor_or~
```

```
30 AUVA    Australian Values Party                                              minor_or~
31 GAP     The Great Australian Party                                           minor_or~
32 KAP     Katter's Australian Party (KAP)                                      minor_or~
33 HMP     Legalise Cannabis Australia                                          minor_or~
34 AUP     Australian Progressives                                              minor_or~
35 XEN     Centre Alliance                                                      minor_or~
36 DPDA    Drew Pavlou Democratic Alliance                                      minor_or~
37 JLN     Jacqui Lambie Network                                                minor_or~
38 TLOC    The Local Party                                                      minor_or~
39 GVIC    The Greens                                                           minor_or~
40 VNS     Victorian Socialists                                                 minor_or~
41 DHJP    Derryn Hinch's Justice Party                                         minor_or~
42 REAS    Reason Australia                                                     minor_or~
43 GRN     The Greens (WA)                                                      minor_or~
44 AUC     Australian Christians                                                minor_or~
45 WAP     WESTERN AUSTRALIA PARTY                                              minor_or~
46 GRN     Australian Greens                                                    minor_or~
47 IMO     HEART Party                                                          minor_or~
48 AJP     AJP                                                                  minor_or~
49 FFPA    Family First                                                         minor_or~
50 CYA     Trumpet of Patriots                                                  minor_or~
51 LTP     Libertarian                                                          minor_or~
52 SOPA    FUSION | Planet Rescue | Whistleblower Protection | Innova~ minor_or~
53 HMP     Legalise Cannabis Party                                              minor_or~
54 GRPF    Gerard Rennick People First                                          minor_or~
55 GRN     The Greens (WA) Inc                                                  minor_or~
```

```r
## Calculate national first preference vote percentages

first_prefs <- data_classified |>
  filter(CountNumber == 0,
         CalculationType == "Preference Count")

# Calculate total votes per party type per year
vote_shares <- first_prefs |>
  group_by(year, PartyType) |>
  summarise(total_votes = sum(CalculationValue, na.rm = TRUE)) |>
  mutate(pct = total_votes / sum(total_votes) * 100) |>
  ungroup()

# View results
vote_shares
```

```
# A tibble: 6 x 4
   year PartyType      total_votes   pct
  <dbl> <chr>                <dbl> <dbl>
1  2022 major_coal         5233334  35.7
2  2022 major_labour       4776030  32.6
3  2022 minor_or_indep     4649678  31.7
4  2025 major_coal         4929402  31.8
5  2025 major_labour       5354138  34.6
6  2025 minor_or_indep     5206696  33.6
```

```
# Save processed data
write_csv(data_classified, "data/data_classified.csv")
```

## Spatial analysis

```
# Calculate vote share per electorate per year

electorate_votes <- first_prefs |>
  group_by(year, DivisionNm, PartyType) |>
  summarise(total_votes = sum(CalculationValue, na.rm = TRUE)) |>
  mutate(pct = total_votes / sum(total_votes) * 100) |>
  ungroup()

electorate_votes
```

```
# A tibble: 903 x 5
    year DivisionNm PartyType      total_votes   pct
   <dbl> <chr>      <chr>                <dbl> <dbl>
 1  2022 Adelaide   major_coal           36080  32.0
 2  2022 Adelaide   major_labour         45086  40.0
 3  2022 Adelaide   minor_or_indep       31598  28.0
 4  2022 Aston      major_coal           42260  43.1
 5  2022 Aston      major_labour         31949  32.5
 6  2022 Aston      minor_or_indep       23951  24.4
 7  2022 Ballarat   major_coal           26142  27.1
 8  2022 Ballarat   major_labour         43171  44.7
 9  2022 Ballarat   minor_or_indep       27187  28.2
10  2022 Banks      major_coal           41622  45.2
# i 893 more rows
```

```r
# Calculate percentage point change between 2022 and 2025

electorate_change <- electorate_votes |>
  select(year, DivisionNm, PartyType, pct) |>
  pivot_wider(names_from = year, values_from = pct, names_prefix = "pct_") |>
  mutate(
    pct_change = pct_2025 - pct_2022,
    trend_dir = case_when(
      pct_change > 0 ~ "increase",
      pct_change < 0 ~ "decrease",
      TRUE ~ "no_change"
    )
  )

# Save processed data
write_csv(electorate_change, "data/electorate_change.csv")

electorate_change
```

```
# A tibble: 456 x 6
   DivisionNm PartyType      pct_2022 pct_2025 pct_change trend_dir
   <chr>      <chr>             <dbl>    <dbl>      <dbl> <chr>
 1 Adelaide   major_coal         32.0     24.2     -7.83  decrease
 2 Adelaide   major_labour       40.0     46.5      6.50  increase
 3 Adelaide   minor_or_indep     28.0     29.3      1.33  increase
 4 Aston      major_coal         43.1     37.7     -5.38  decrease
 5 Aston      major_labour       32.5     37.3      4.71  increase
 6 Aston      minor_or_indep     24.4     25.1      0.672 increase
 7 Ballarat   major_coal         27.1     28.6      1.50  increase
 8 Ballarat   major_labour       44.7     42.4     -2.39  decrease
 9 Ballarat   minor_or_indep     28.2     29.1      0.891 increase
10 Banks      major_coal         45.2     39.1     -6.12  decrease
# i 446 more rows
```

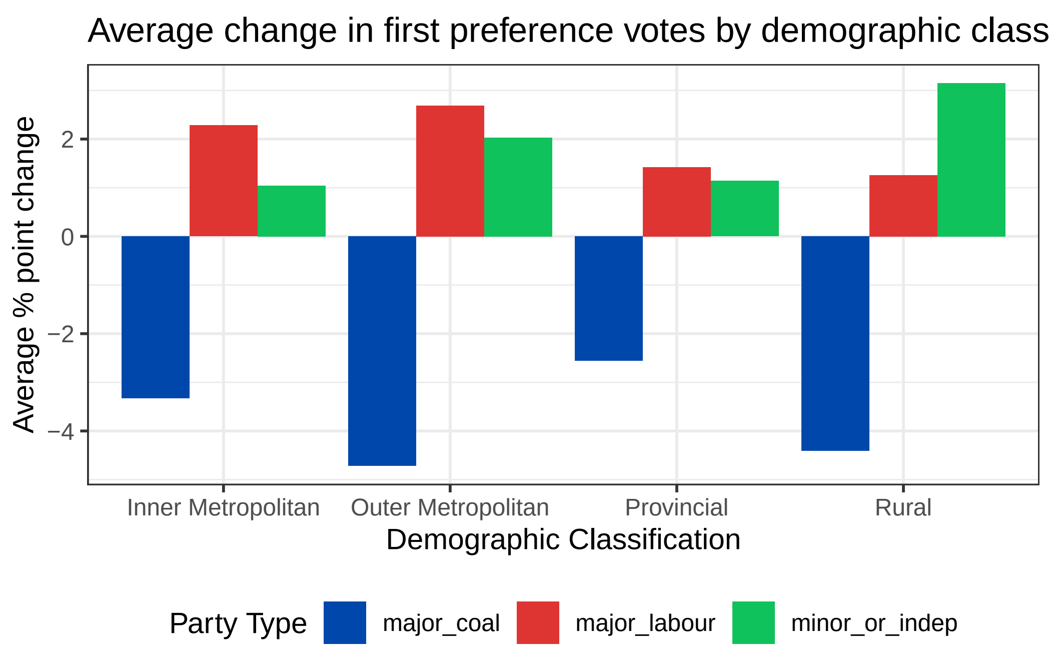
```r
demographic <- read_csv(
  "data/demographic-classification-as-at-4-march-2025.csv",
  skip = 3
) |>
  select(1:3) |>
  rename(
    DivisionNm = 1,
```

```r
    DemographicClass = 2,
    State = 3
  )

# Join demographic classifications to electorate change data
electorate_change_demo <- electorate_change |>
  left_join(demographic, by = "DivisionNm")

# Summarise average pct_change by demographic class and party type
demo_summary <- electorate_change_demo |>
  filter(!is.na(DemographicClass)) |>
  group_by(DemographicClass, PartyType) |>
  summarise(avg_pct_change = mean(pct_change, na.rm = TRUE)) |>
  ungroup()

demo_summary
```

```
# A tibble: 12 x 3
   DemographicClass   PartyType      avg_pct_change
   <chr>              <chr>                   <dbl>
 1 Inner Metropolitan major_coal              -3.32
 2 Inner Metropolitan major_labour             2.28
 3 Inner Metropolitan minor_or_indep           1.04
 4 Outer Metropolitan major_coal              -4.72
 5 Outer Metropolitan major_labour             2.69
 6 Outer Metropolitan minor_or_indep           2.03
 7 Provincial         major_coal              -2.56
 8 Provincial         major_labour             1.42
 9 Provincial         minor_or_indep           1.14
10 Rural              major_coal              -4.41
11 Rural              major_labour             1.26
12 Rural              minor_or_indep           3.15
```

```r
# Bar chart visualisation for demographic summary
ggplot(demo_summary, aes(x = DemographicClass, y = avg_pct_change, fill = PartyType)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c(
    "major_coal" = "#0047AB",
    "major_labour" = "#DE3533",
    "minor_or_indep" = "#10C25B"
  )) +
```

```r
  labs(
    title = "Average change in first preference votes by demographic classification",
    x = "Demographic Classification",
    y = "Average % point change",
    fill = "Party Type"
  ) +
  theme_bw() +
  theme(legend.position = "bottom")
```




## Demographic deep dive

```r
## Selecting an Electorate with Significant Voting Shift
# Exploratory analysis

electorate_change |>
  filter(PartyType == "minor_or_indep") |>
  arrange(desc(pct_change)) |>
  print(n = 20)
```

```
# A tibble: 152 x 6
```

```
   DivisionNm PartyType      pct_2022 pct_2025 pct_change trend_dir
   <chr>      <chr>             <dbl>    <dbl>      <dbl> <chr>
 1 Calare     minor_or_indep     37.2     59.9      22.7  increase
 2 Calwell    minor_or_indep     31.5     53.8      22.3  increase
 3 Blaxland   minor_or_indep     18.0     34.4      16.4  increase
 4 Watson     minor_or_indep     21.6     36.8      15.2  increase
 5 Paterson   minor_or_indep     22.6     35.7      13.1  increase
 6 Farrer     minor_or_indep     28.8     41.5      12.7  increase
 7 Forrest    minor_or_indep     29.4     41.5      12.0  increase
 8 Monash     minor_or_indep     36.6     47.9      11.3  increase
 9 Fremantle  minor_or_indep     31.8     42.5      10.7  increase
10 Moore      minor_or_indep     25.4     36.0      10.6  increase
11 Parkes     minor_or_indep     30.5     40.4       9.91 increase
12 Wannon     minor_or_indep     36.4     45.8       9.36 increase
13 Lyne       minor_or_indep     35.0     43.9       8.92 increase
14 Burt       minor_or_indep     25.4     34.1       8.71 increase
15 Riverina   minor_or_indep     33.0     41.2       8.29 increase
16 Fisher     minor_or_indep     32.4     40.5       8.09 increase
17 Flynn      minor_or_indep     29.6     37.5       7.93 increase
18 Wills      minor_or_indep     43.8     51.5       7.71 increase
19 Makin      minor_or_indep     22.3     29.7       7.43 increase
20 Bean       minor_or_indep     28.6     35.9       7.36 increase
# i 132 more rows
```

```
electorate_change |>
  filter(PartyType == "major_coal") |>
  arrange(pct_change) |>
  print(n = 20)
```

```
# A tibble: 152 x 6
   DivisionNm PartyType  pct_2022 pct_2025 pct_change trend_dir
   <chr>      <chr>         <dbl>    <dbl>      <dbl> <chr>
 1 Calare     major_coal     47.7     29.7     -18.0  decrease
 2 Braddon    major_coal     44.1     31.7     -12.4  decrease
 3 Watson     major_coal     26.5     15.2     -11.4  decrease
 4 Hasluck    major_coal     33.4     22.2     -11.2  decrease
 5 Lyons      major_coal     37.2     26.2     -11.0  decrease
 6 Moore      major_coal     41.8     31.4     -10.4  decrease
 7 Hindmarsh  major_coal     32.7     23.1      -9.59 decrease
 8 Robertson  major_coal     40.0     30.4      -9.58 decrease
 9 Paterson   major_coal     36.7     27.2      -9.45 decrease
10 Leichhardt major_coal     36.7     27.3      -9.44 decrease
```

```
11 Parkes     major_coal     49.3     40.0      -9.36 decrease
12 Bonner     major_coal     44.8     35.5      -9.34 decrease
13 Rankin     major_coal     29.0     19.7      -9.26 decrease
14 Hunter     major_coal     27.4     18.2      -9.25 decrease
15 Wright     major_coal     43.2     34.1      -9.13 decrease
16 Makin      major_coal     31.4     22.5      -8.95 decrease
17 Farrer     major_coal     52.3     43.4      -8.85 decrease
18 Sturt      major_coal     43.1     34.3      -8.80 decrease
19 Bass       major_coal     39.7     31.4      -8.29 decrease
20 Calwell    major_coal     23.7     15.7      -7.96 decrease
# i 132 more rows
```

```
electorate_change |>
  filter(DivisionNm == "Monash", PartyType == "major_coal")
```

```
# A tibble: 1 x 6
  DivisionNm PartyType  pct_2022 pct_2025 pct_change trend_dir
  <chr>      <chr>         <dbl>    <dbl>      <dbl> <chr>
1 Monash     major_coal     37.8     31.8      -6.01 decrease
```

```
census_g02 <- read_csv("data/2021Census_G02_VIC_SA1.csv")
glimpse(census_g02)
```

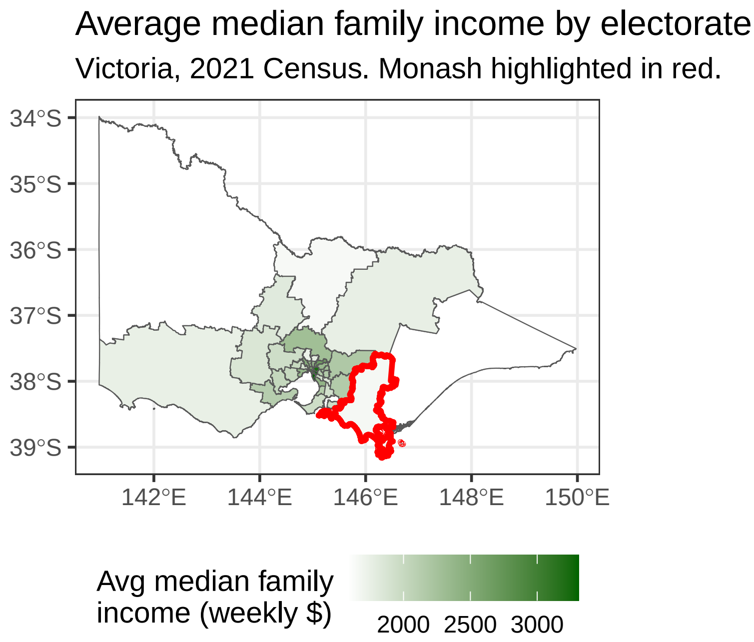
```
Rows: 15,482
Columns: 9
$ SA1_CODE_2021                <dbl> 20101100101, 20101100102, 20101100105, 2~
$ Median_age_persons           <dbl> 39, 49, 33, 40, 46, 35, 39, 45, 40, 47, ~
$ Median_mortgage_repay_monthly <dbl> 1452, 1257, 1136, 1302, 1044, 1517, 1300~
$ Median_tot_prsnl_inc_weekly  <dbl> 713, 629, 552, 642, 623, 840, 733, 857, ~
$ Median_rent_weekly           <dbl> 330, 235, 279, 330, 280, 355, 280, 400, ~
$ Median_tot_fam_inc_weekly    <dbl> 1852, 2050, 1266, 1562, 1212, 2111, 1483~
$ Average_num_psns_per_bedroom <dbl> 0.8, 0.7, 0.8, 0.7, 0.7, 0.8, 0.7, 0.8, ~
$ Median_tot_hhd_inc_weekly    <dbl> 1224, 1281, 954, 1271, 891, 1858, 1184, ~
$ Average_household_size       <dbl> 2.6, 2.4, 2.0, 2.5, 1.8, 2.7, 2.2, 3.1, ~
```

```
## Analyse spatial distribution of demographic characteristics

# Load the maps
aec_map <- read_sf("data/vic-july-2021-esri/E_VIC21_region.shp") |>
  mutate(Elect_div = toupper(Elect_div))
```

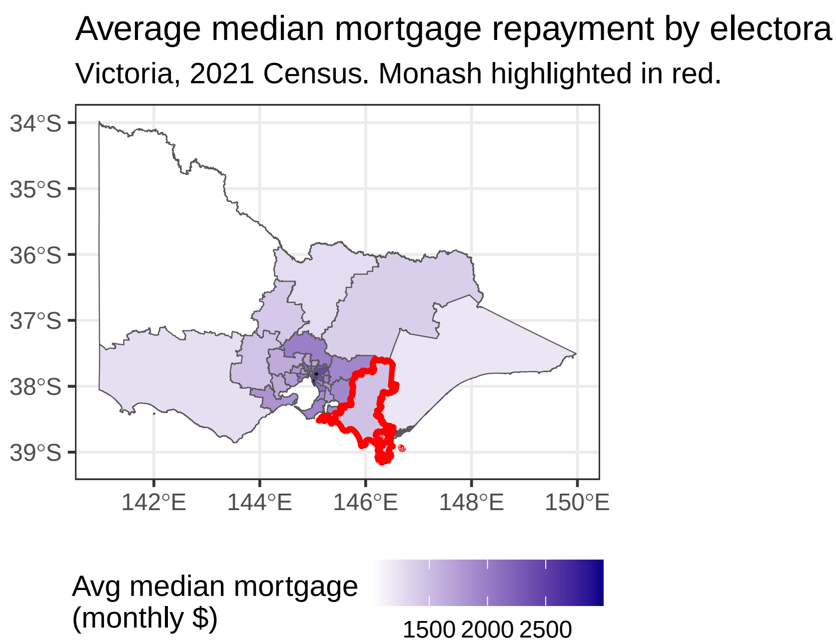
```r
vicmap_sa1_G02 <- read_sf("data/Geopackage_2021_G02_VIC_GDA2020/G02_VIC_GDA2020.gpkg",
                           layer = "G02_SA1_2021_VIC")

# Transform projection
aec_map$geometry = st_transform(aec_map$geometry, 7844)


# Get SA1 centroids
vicmap_sa1_G02_w_centroid <- vicmap_sa1_G02 |>
  mutate(centroid = st_centroid(geom))

# Find which electorate each SA1 belongs to
electoral_intersects = st_intersects(vicmap_sa1_G02_w_centroid$centroid,
                                     aec_map$geometry,
                                     sparse = FALSE)

arr_ind = which(electoral_intersects == TRUE, arr.ind = TRUE)

sa1_ind = arr_ind[,1]
division_ind = arr_ind[,2]
division_name = aec_map$Elect_div[division_ind]
sa1_name = vicmap_sa1_G02_w_centroid$SA1_NAME_2021[sa1_ind]

# Save SA1 to electorate mapping
sa1_divisions = data.frame(SA1_NAME_2021 = sa1_name,
                           DivisionNm = division_name)

# Add electorate names to census data
vicmap_sa1_G02_electorates <- vicmap_sa1_G02 |>
  right_join(sa1_divisions)

# Group by electorate and calculate averages
vicmap_sa1_G02_electorates_grouped <- vicmap_sa1_G02_electorates |>
  group_by(DivisionNm) |>
  summarise(
    mean_fam_income = mean(Median_tot_fam_inc_weekly, na.rm = TRUE),
    mean_mortgage = mean(Median_mortgage_repay_monthly, na.rm = TRUE)
  ) |>
  ungroup()

head(vicmap_sa1_G02_electorates_grouped)
```

```
Simple feature collection with 6 features and 3 fields
Geometry type: POLYGON
Dimension:     XY
Bounding box:  xmin: 143.4406 ymin: -38.0745 xmax: 146.1925 ymax: -36.37084
Geodetic CRS:  GDA2020
# A tibble: 6 x 4
  DivisionNm mean_fam_income mean_mortgage                                  geom
  <chr>                <dbl>         <dbl>                        <POLYGON [°]>
1 ASTON                2164.         1953. ((145.2593 -37.83661, 145.2591 -
37.8~
2 BALLARAT             1862.         1460. ((143.6553 -37.60974, 143.6545 -
37.6~
3 BENDIGO              1814.         1423. ((144.518 -36.72529, 144.5181 -
36.72~
4 BRUCE                1840.         1755. ((145.2192 -37.94013, 145.2191 -
37.9~
5 CALWELL              1732.         1693. ((144.8676 -37.67103, 144.8676 -
37.6~
6 CASEY                2181.         1925. ((145.3088 -37.79399, 145.3088 -
37.7~
```

```r
# Create a separate dataset for Monash only
monash_only <- vicmap_sa1_G02_electorates_grouped |>
  filter(DivisionNm == "MONASH")


# Plot 1 - median family income by electorate
ggplot(vicmap_sa1_G02_electorates_grouped) +
  geom_sf(aes(geometry = geom, fill = mean_fam_income)) +
  geom_sf(data = monash_only, aes(geometry = geom),
          fill = NA, color = "red", linewidth = 1) +
  scale_fill_gradient(low = "white", high = "darkgreen") +
  labs(title = "Average median family income by electorate",
       subtitle = "Victoria, 2021 Census. Monash highlighted in red.",
       fill = "Avg median family\nincome (weekly $)") +
  theme_bw() +
  theme(legend.position = "bottom")
```




```
# Plot 2 - median mortgage repayment by electorate
ggplot(vicmap_sa1_G02_electorates_grouped) +
  geom_sf(aes(geometry = geom, fill = mean_mortgage)) +
  geom_sf(data = monash_only, aes(geometry = geom),
          fill = NA, color = "red", linewidth = 1) +
  scale_fill_gradient(low = "white", high = "darkblue") +
  labs(title = "Average median mortgage repayment by electorate",
       subtitle = "Victoria, 2021 Census. Monash highlighted in red.",
       fill = "Avg median mortgage\n(monthly $)") +
  theme_bw() +
  theme(legend.position = "bottom")
```




```r
## Quality checks on spatial join

# How many SA1s didn't match any electorate?
nrow(vicmap_sa1_G02) - nrow(vicmap_sa1_G02_electorates)
```

```
[1] 9
```

```r
# Any missing values in our key variables?
vicmap_sa1_G02_electorates_grouped |>
  summarise(
    missing_income = sum(is.na(mean_fam_income)),
    missing_mortgage = sum(is.na(mean_mortgage))
  )
```

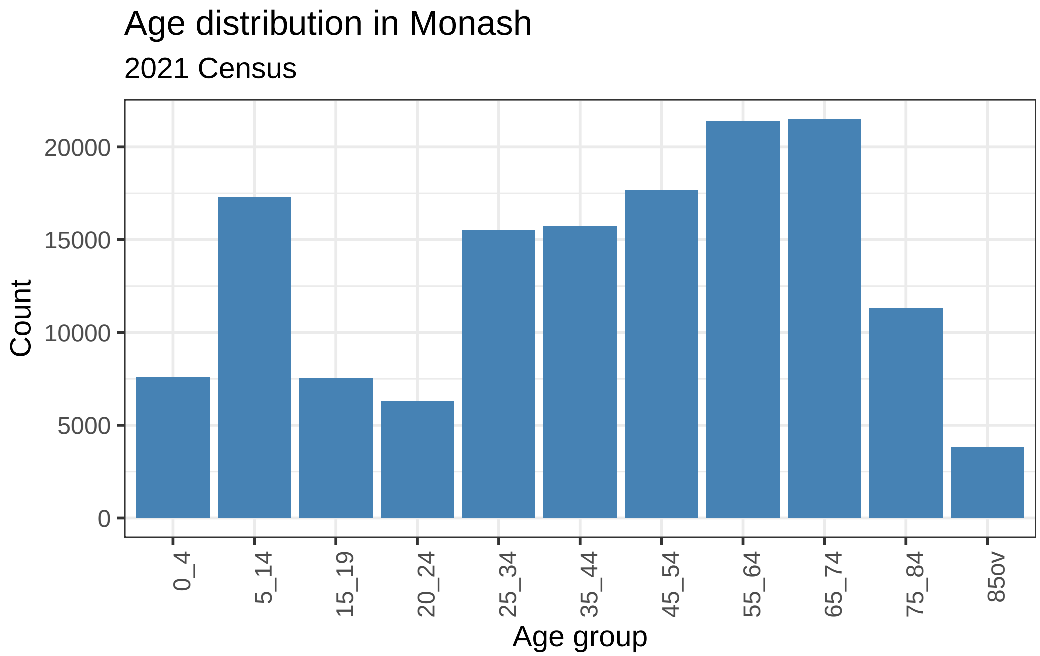
```
Simple feature collection with 1 feature and 2 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: 140.9619 ymin: -39.15918 xmax: 149.9762 ymax: -33.98064
Geodetic CRS:  GDA2020
# A tibble: 1 x 3
  missing_income missing_mortgage                                       geom
           <int>            <int>                         <MULTIPOLYGON [°]>
1              0                0 (((144.6863 -38.24588, 144.6869 -38.24571, 14~
```

```r
# Compare Monash vs Rest of VIC for census variables
vicmap_sa1_G02_electorates_grouped |>
  mutate(is_monash = ifelse(DivisionNm == "MONASH", "Monash", "Rest of VIC")) |>
  group_by(is_monash) |>
  summarise(
    avg_fam_income = mean(mean_fam_income, na.rm = TRUE),
    avg_mortgage = mean(mean_mortgage, na.rm = TRUE)
  )
```

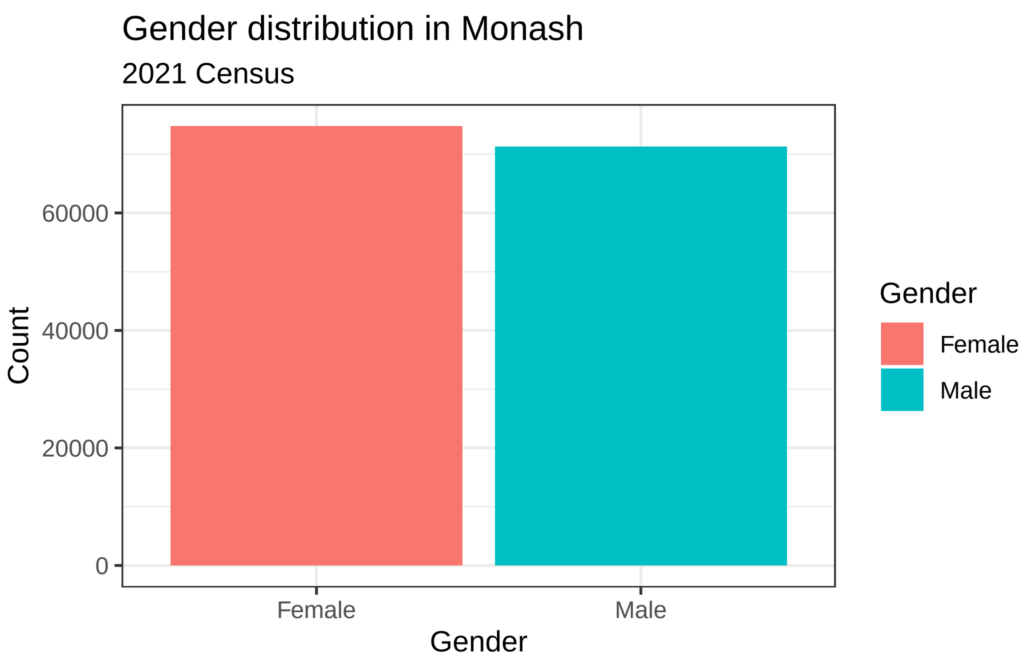
```
Simple feature collection with 2 features and 3 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: 140.9619 ymin: -39.15918 xmax: 149.9762 ymax: -33.98064
Geodetic CRS:  GDA2020
# A tibble: 2 x 4
  is_monash   avg_fam_income avg_mortgage                                   geom
  <chr>                <dbl>        <dbl>                     <MULTIPOLYGON [°]>
1 Monash               1660.        1470. (((145.2893 -38.45516, 145.2896 -
38.4~
2 Rest of VIC          2184.        1908. (((146.1919 -37.58752, 146.1923 -
37.5~
```

```r
## Explore demographic profile of selected electorate

# Load libraries
library(tidyverse)
library(sf)

# Load data
g01 <- read_csv("data/2021Census_G01_VIC_SA1.csv") |>
  mutate(SA1_CODE_2021 = as.character(SA1_CODE_2021)) |>
  left_join(sa1_divisions, by = c("SA1_CODE_2021" = "SA1_NAME_2021")) |>
  mutate(region = ifelse(DivisionNm == "MONASH", "Monash", "Rest of VIC"))

g17a <- read_csv("data/2021Census_G17A_VIC_SA1.csv")


# Tidy Age and Gender data
g01_long <- g01 |>
  select(SA1_CODE_2021, region,
         starts_with("Age_0_4"), starts_with("Age_5_14"),
         starts_with("Age_15_19"), starts_with("Age_20_24"),
```

```
         starts_with("Age_25_34"), starts_with("Age_35_44"),
         starts_with("Age_45_54"), starts_with("Age_55_64"),
         starts_with("Age_65_74"), starts_with("Age_75_84"),
         starts_with("Age_85ov"),
         -ends_with("_P")) |>
  pivot_longer(cols = -c(SA1_CODE_2021, region),
               names_to = "category",
               values_to = "count") |>
  mutate(
    sex = str_extract(category, "[MF]$"),
    age = str_remove(category, "_[MF]$") |>
          str_remove("Age_") |>
          str_remove("_yr")
  )


# PLOT 1 - AGE DISTRIBUTION
g01_long |>
  filter(region == "Monash") |>
  group_by(age) |>
  summarise(total = sum(count, na.rm = TRUE)) |>
  ungroup() |>
  mutate(age = fct_relevel(age, "0_4", "5_14", "15_19", "20_24",
                           "25_34", "35_44", "45_54", "55_64",
                           "65_74", "75_84", "85ov")) |>
  ggplot(aes(x = age, y = total)) +
  geom_col(fill = "steel blue") +
  labs(title = "Age distribution in Monash",
       subtitle = "2021 Census",
       x = "Age group", y = "Count") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```

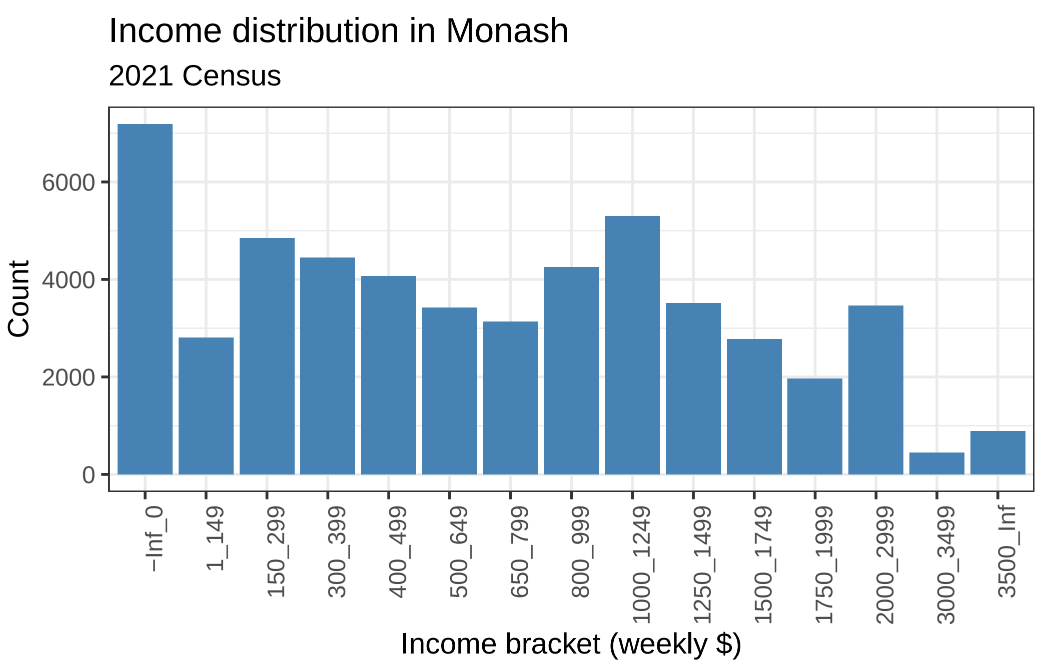



```r
# PLOT 2 - GENDER DISTRIBUTION
g01 |>
  filter(region == "Monash") |>
  summarise(Male = sum(Tot_P_M, na.rm = TRUE),
            Female = sum(Tot_P_F, na.rm = TRUE)) |>
  pivot_longer(cols = everything(),
               names_to = "gender",
               values_to = "count") |>
  ggplot(aes(x = gender, y = count, fill = gender)) +
  geom_col() +
  scale_color_viridis_c() +
  labs(title = "Gender distribution in Monash",
       subtitle = "2021 Census",
       x = "Gender", y = "Count", fill = "Gender") +
  theme_bw()
```




```r
# Tidy income data
g17_long <- g17a |>
  pivot_longer(cols = -1, names_to = "category", values_to = "count") |>
  filter(!str_detect(category, "Tot"),
         !str_detect(category, "PI_NS")) |>
  mutate(
    category = str_replace(category, "Neg_Nil_income", "-Inf_0"),
    category = str_replace(category, "Neg_Nil_incme", "-Inf_0"),
    category = str_replace(category, "Negtve_Nil_incme", "-Inf_0"),
    category = str_replace(category, "more", "Inf"),
    category = str_replace(category, "85ov", "85_110_yrs"),
    category = str_replace(category, "85_yrs_ovr", "85_110_yrs")
  ) |>
  separate_wider_delim(category, delim = "_",
                       names = c("sex", "income_min", "income_max", "age_min", "age_max", "e
                       too_many = "drop",
                       too_few = "align_start") |>
  mutate(
    income_bracket = paste(income_min, income_max, sep = "_"),
    SA1_CODE_2021 = as.character(SA1_CODE_2021)
  ) |>
  left_join(sa1_divisions, by = c("SA1_CODE_2021" = "SA1_NAME_2021")) |>
  mutate(region = ifelse(DivisionNm == "MONASH", "Monash", "Rest of VIC"))
```

```r
# PLOT 3 - INCOME DISTRIBUTION
g17_long |>
  filter(region == "Monash",
         sex %in% c("M", "F")) |>
  group_by(income_bracket, income_min) |>
  summarise(total = sum(count, na.rm = TRUE)) |>
  ungroup() |>
  mutate(income_min = as.numeric(income_min),
         income_bracket = fct_reorder(income_bracket, income_min)) |>
  ggplot(aes(x = income_bracket, y = total)) +
  geom_col(fill = "steel blue") +
  labs(title = "Income distribution in Monash",
       subtitle = "2021 Census",
       x = "Income bracket (weekly $)", y = "Count") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
```



# Key Findings

### Fact Checker method and results

### Method:

To fact-check the claim made by the Australia Institute, I gathered the distribution of the first preference votes by candidates and divisions for the 2022 and 2025 federal elections. I put these two sets of data together into a single dataframe along with a new year variable indicating which year an entry belongs to. Afterwards, I created a party category which then classified each party into one of three groups using case_when() in R:

- major_labour – Australian Labor Party (ALP)
- major_coal – Liberal Party (LP), National Party (NP), Liberal National Party (LNP), and Country Liberal Party (CLP)
- minor_or_indep – any other parties and independents (including Greens and One Nation)

The calculation for the national vote share of the first preference was performed by filtering first preference votes (CountNumber == 0 & CalculationType == "Preference Count") and summing up the votes by PartyType within years to compute the proportion.

The AEC election data is publicly available and free to use under the Commonwealth of Australia copyright. The ABS Census data is similarly available under Creative Commons Attribution 4.0 International licence (CC BY 4.0).

### Results:

My estimated Coalition first preference vote for 2025 was 31.8% which is almost similar but not fully equal to the 32% obtained by the Australia Institute's report. The difference could be because of rounding, which was done in the article. It is clear that the claim made by the Australia Institute is correct. In 2025, the minor parties and independents have more first preference votes compared to the coalition for the first time in history; this translates to 33.6% compared to 31.8%. This is an unprecedented change considering that in 2022, the minor parties and independents had 31.7% while the coalition had 35.7%.

## Spatial Analysis method and results

### Method

To see whether there was a consistency in the pattern observed at the national level spatially, the change in vote share at the electorate level was compared against the demographic classification of the AEC (as at 4 March 2025). Each electorate is classified according to one of the following categories: Inner Metropolitan, Outer Metropolitan, Provincial, and Rural.

The independent/minor party votes in some electorates could potentially be zero in either or both years since no independent/minor party candidate contested. Such electorates could be in Rural or Provincial electorates. This could result in overstating or understating the percentage points gained by major parties depending on whether there were no independent/minor parties in either of the two years.

### Results

The bar graph indicates the average percentage change in the share of first-preference votes between 2022 and 2025, for each demographic classification. As shown in the bar chart below, the national trend was seen in all four demographic classifications:

- The Coalition lost ground in all four classifications, but the biggest decline occurred in Outer Metropolitan seats (-4.7%), while the least decline occurred in Provincial seats (-2.6%)
- Labor saw an increase in all four demographic classifications, but the largest gain came in Outer Metropolitan seats (+2.7%)
- Minor parties and independents showed an increase in all four demographic classifications, with the biggest increase occurring in Rural seats (+3.1%)

It can be concluded from these results that the national pattern is not specific to any type of electorate but rather represents a pattern in individual electorates.

## Demographic Deep Dive and Results

The electorate of Monash was chosen to be studied due to its strong swing towards independents and minor parties (+11.3%), along with an evident swing against the Coalition Party (-6.0%) within the time frame of 2022-2025. The electorate of Monash is located in Melbourne's outer eastern suburbs.

### Census Variables

Two variables were selected from the 2021 ABS Census G02 dataset:

- Median weekly family income (`Median_tot_fam_inc_weekly`) - Historically wealthier communities supported the Coalition, making income an interesting variable to examine given the swing away from Coalition in electorates like Monash.
- Median monthly mortgage repayment (`Median_mortgage_repay_monthly`) — Housing affordability was one of the key issues in the 2025 election. Monthly mortgage repayments include both the stress related to paying off a loan and the value of property owned, which makes this particular variable different from just income measures. Higher mortgage payments usually correspond to high property prices.

### Spatial Join Method

Census SA1 data was spatially joined to AEC electoral boundaries using the centroid method, where SA1 centroids were intersected with electorate boundary polygons. The VIC 2021 AEC boundaries were reprojected from GDA1994 to GDA2020 (EPSG:7844) to match the census spatial data.

### Quality Checks and Limitations

- Nine SA1s could not be matched to any electorate, likely because their centroids fell on boundaries or in water areas. No missing values were found for either census variable after joining.
- The census data is from 2021 while the electoral data is from 2025, a four year gap during which Monash's demographic profile may have changed. The census should therefore be treated as an approximation of the 2025 voter demographic.
- Aggregating SA1 level medians by taking the mean of medians is a methodological simplification that may slightly over or underestimate true electorate level values.

### Results

Comparing Monash to the rest of Victoria:

| Area | Avg median family income (weekly) | Avg median mortgage (monthly) |
|---|---|---|
| Monash | \$1,660 | \$1,470 |
| Rest of VIC | \$2,184 | \$1,908 |

Surprisingly, Monash scores even below the state average for both of these indicators, especially relative to inner Melbourne seats, indicating that it is not a typical high-income or high-mortgage electorate.

The demographic plots reveal the following about Monash's voter profile:

- Age - The age distribution in Monash shows a notably older profile, with the largest groups being 55-64 and 65-74 year olds. Younger age groups (20-34) are underrepresented compared to older cohorts.
- Gender - The gender distribution in Monash electorate reflects a fairly even ratio between males and females in Monash – similar to the state norm.
- Income - The income distribution in Monash is concentrated within the lower-middle income levels and very few people belong to the higher income classes.

Monash is broadly representative of Victoria in terms of gender distribution and age profile. But it falls below average with regard to family income levels and home mortgage payments as compared with other Victorian electorates. This makes the swing toward independents particularly interesting, driven not by wealthy inner-city Teal voters but by older, middle-income outer suburban residents.

## CITATIONS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